



Paper

Platform for precise, personalised glucose forecasting through continuous glucose and physical activity monitoring and deep learning[☆]

Deepjyoti Kalita^{a,*}, Hrishita Sharma^a, Jayanta Kumar Panda^b, Khalid B. Mirza^{a,**}

^a National Institute of Technology, Rourkela, Sundargarh, 769008, Odisha, India

^b SCB Medical College and Hospital, Cuttack, 753001, Odisha, India



ARTICLE INFO

Keywords:

Closed loop diabetes management
Artificial pancreas
Deep learning
Content-based attention learning
Glucose forecasting

ABSTRACT

Emerging research has demonstrated the advantage of continuous glucose monitoring for use in artificial pancreas and diabetes management in general. Recent studies demonstrate that glucose level forecasting using deep learning can help avoid postprandial hyperglycemia (≥ 180 mg/dL) or hypoglycemia (≤ 70 mg/dL) from delayed or increased insulin dosing in artificial pancreas. In this paper, a novel hybrid deep learning framework with integration of content-based attention learning is presented, to effectively predict the glucose measurements with prediction horizons (PH) = 15, 30 and, 60 minutes for T1D and T2D patients based on past data. We also present a complete cloud-based system and mobile app used for collecting CGM sensor, physical activity data, CHO values and insulin measurements to perform glucose forecasts using the proposed model running on Cloud. This model was validated using clinical data of individual with Type 1 diabetes (OhioT1DM) and individual with Type 2 diabetes. The mean absolute relative difference (MARD) was 12.33 ± 3.15 , $7.14 \pm 1.76\%$ for PH = 60 and, 30 min respectively on OhioT1DM clinical Dataset. The root mean squared error (RMSE) was 29.41 ± 5.92 mg/dL and 17.19 ± 3.22 mg/dL and the mean absolute error (MAE) was 21.96 ± 4.67 mg/dL and 12.58 ± 2.34 mg/dL for PH = 60 and, 30 min respectively on the same clinical dataset. It was observed that inclusion of physical activity leads to improved glucose forecasting accuracy. Furthermore, all these results were obtained by training the model on only 8 days of clinical data of a single patient, followed by testing on clinical data on the following days. The results indicate that training on a single patient data may lead to better *personalisation* and better glucose forecasting results compared to existing works.

1. Introduction

Diabetes, especially in individual with Type 1 diabetes, is a chronic disease that impairs the body's ability to maintain blood glucose homeostasis. α - cells and β - cells in the pancreas govern the supply of hormones such as glucagon and insulin respectively, to maintain appropriate glucose levels [1]. Individual with Type 2 diabetes (T2D) and individual with Type 1 diabetes (T1D) are the most prevalent Types of diabetes. Blood glucose homeostasis is maintained through exogenous insulin as β -cells are inadequate to generate the necessary insulin in this condition [2–4].

Glucose monitoring and measurement is required as part of the traditional approach to diabetes care. Patients need to actively take blood glucose measurements multiple times daily. Continuous glucose mon-

itoring (CGM) sensors, allow for a higher rate of automated glucose sampling every 5 - 15 minutes [5], depending on the type of sensor. This approach improves the results of clinical trial participants by controlling blood sugar levels [6,7]. Forecasting glucose levels and insulin dosage, enables users to preemptively plan physical activities in order to minimize the likelihood of unfavourable hypoglycaemic events and can help enhance glucose management even further. Multiple variables impact glucose fluctuation, such as variability in insulin dosage, physical exercise, psychological stress and changes in meal intake quantity posing problems [8]. Furthermore, significant subject variability influences individual glycaemic reactions, resulting in disparities among individual patients [9,10].

Few studies [11–19] have utilised deep learning to predict glucose levels for individual with Type 1 diabetes. However, previous studies

[☆] This work was funded by Indian Council of Medical Research under grant no. IIRP/23-24/7414.

^{*} Corresponding author.

^{**} Principal corresponding author.

E-mail addresses: deepjyoti_kalita@nitrkl.ac.in (D. Kalita), baigm@nitrkl.ac.in (K.B. Mirza).

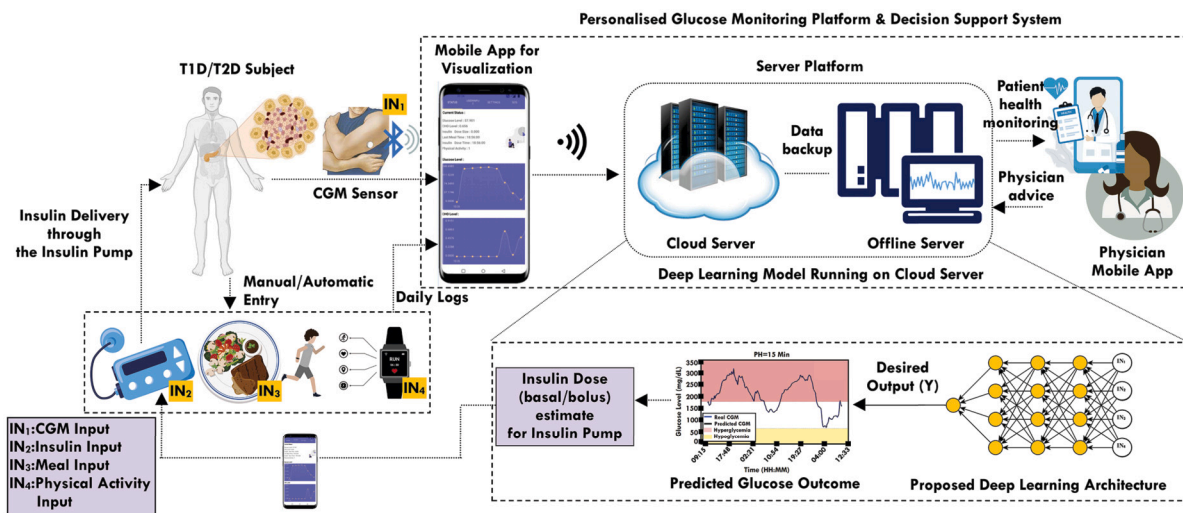


Fig. 1. The overall Diabetes Management Platform is shown here with multiple different physiological inputs. It comprises three main subsystems: Personalised Glucose Monitoring Platform & Decision Support System which is capable of delivering instantaneous readings, forecasting blood glucose levels, and issuing alerts for potential hypoglycemia, dual-server architecture with integration of the deep learning model on the cloud server and different physiological inputs parameter integration under one platform.

lack in either consideration towards physical activity or require large amount of historical training data to work efficiently [20,21]. The platform proposed in this paper provides a complete solution with a novel, deep learning mode that can be trained on small datasets of a single person leading to better forecasting results compared to existing work. The proposed overall diabetes management platform is shown in Fig. 1. They used traditional dense neural networks in their research, but they didn't make use of the benefits of deep layers. A two-layer artificial neural network (ANN) was used in [22] to predict hypo or hyperglycemia. One of the benefits of ANN is that it usually eliminates the need for manual feature discovery. However, in the literature, ANNs are often constructed with fewer than three layers, limiting their learning potential owing to model complexity.

The proposed deep learning framework uses a novel, wide-deep combination of Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) layers to predict CGM values (Prediction Horizon [PH] = 15, 30 and 60 minutes) utilizing past CGM levels, meal intake data, physical activity data and insulin values. The overall framework is shown in Fig. 3 with multiple different physiological inputs. These data include previous glucose values from the CGM sensor (*CGM*), meal quantity (*CHO*) and insulin dose (*Insulin*) data. The time series data of *CGM*, *Insulin*, *Physical activity data* and *CHO* are fed input to the deep learning framework for training. After validation with a clinical dataset, as a test case, the proposed architecture is also tested live with T2D clinical data of individual with Type 2 diabetes.

This framework with the model provides a personalized glucose forecast, with the model evolving as the training dataset changes. Since our model performs better with fewer training data (we use 8 days of training data), it can be more personalised than existing state-of-the-art technology. The suggested approaches’ performance is assessed on different datasets of simulated instances of T1DM participants using the FDA-approved UVA/Padova simulator [23], OhioT1D data set and locally collected T2D data set and compared to other glucose prediction algorithms. The findings demonstrate the method’s competitive performance.

To the best of our knowledge, the method proposed for glucose measurement forecasting, is the first that uses and explores the use of hybrid LSTM and GRU layers for glucose forecasting using wide-deep based architecture, while incorporating seamless integration of physical activity metrics from wearable physical activity monitoring devices. It outperforms benchmark algorithms in terms of Mean absolute relative difference (MARD), Mean absolute error (MAE), and Root mean square

error (RMSE) and obtains competitive forecast performance when compared to them. A very preliminary version of this work, consisting of the network with limited inputs was presented in [24].

In this paper, Section 2 presents prior work in the area of glucose prediction, followed by a description of the system architecture and experimental methods in Section 3. It also contains in-depth description regarding the proposed framework. In Section 4, the performance metrics are examined and analysed for the proposed framework. The method’s main aspects are discussed in 5. Section 6 concludes the paper.

2. Background

2.1. Impact of lag on glucose sensors measurement and need of glucose forecasting

Commercially available glucose sensors use a transcutaneous electrode to find glucose in the interstitial fluid (ISF). When calibrating the sensor signal to blood glucose (BG) levels, capillary blood is measured using a handheld metre [25]. Studies support the accuracy of subcutaneous (SC) glucose sensors calibrated to BG readings across various physiological ranges, in spite of the difficulty in establishing the gradient between absolute blood and ISF glucose concentrations (reported as 50-100%) [25]. When calibrated correctly, regardless of the actual gradient, the sensor measurement in mg/dL equals BG at steady state, assuming a proportional connection.

The difference between the true and measured BG values is inaccurate due to a dynamic lag in the measurement process. The sensor responsiveness to a ramp input yields one measure of this inaccuracy. As a function of the glucose concentration in the major blood arteries, the first-order equation for ISF glucose concentration can be expressed by [25]:

$$\frac{\partial y_m(t)}{\partial t} = -(k_3 + k_2)y_m(t) + k_1 \frac{V_b}{V_{isf}} y_{iv}(t) \quad (1)$$

where $y_m(t)$ is denoted as the blood glucose level as determined by the sensor (measured in mg/dL) and similarly $y_{iv}(t)$ is the blood glucose level (measured in mg/dL), and k_1 , k_2 , and k_3 are the different rate constants. The selection of the ramp, intended to have the maximum expected rate of change for the process, results in an upper bound on the error determined by the equation below-

$$\epsilon_{max} = |y_m(t) - y_{iv}(t)|_{max} / \quad (2)$$

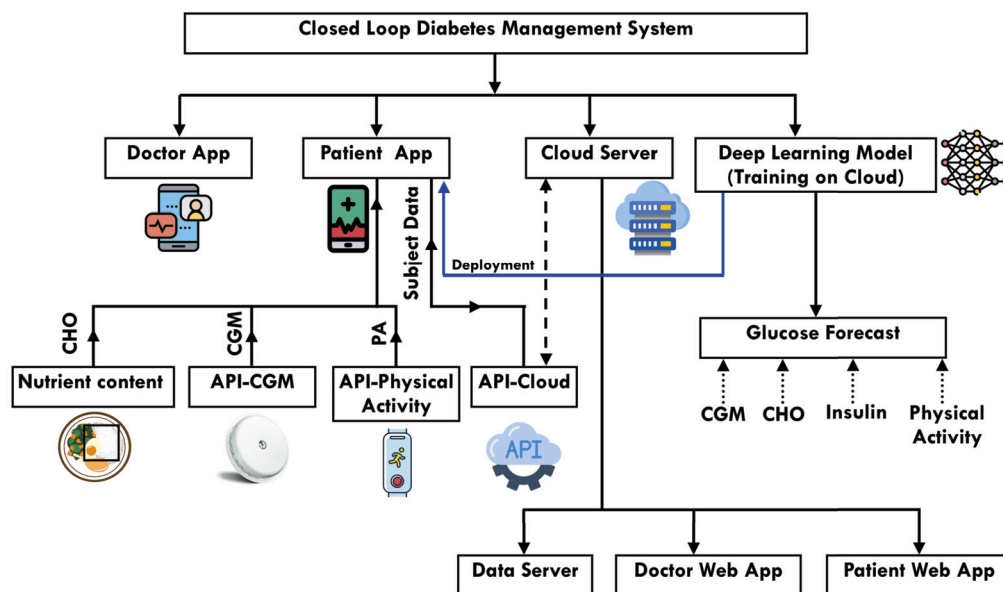


Fig. 2. This figure shows the overall architecture of cloud based patient monitoring system for closed loop diabetes management with incorporation of physical activity.

Where, $y_{iv}(t) = at$ denotes the ramp input for the glucose concentration. In order to determine the time-domain response to a ramp for a transfer function with first-order dynamics and unity gain, the Laplace transform value of this ramp is utilized in the Laplace transform of Eq (1) and then the inverse Laplace transform can be expressed as-

$$y_m(t) = \alpha \tau_s (e^{-\frac{t}{\tau_s}} - 1) + \alpha t \quad (3)$$

After replacing the Eq (2) by Eq (3) can be expressed as

$$\epsilon_{max} = \left| \alpha \tau_s (e^{-\frac{t}{\tau_s}} - 1) \right|_{max} \quad (4)$$

Then the maximum errors produced $t \gg \tau_s$ and expressed as

$$\epsilon_{max} = |\alpha \tau_s| \quad (5)$$

According to clinical data, the greatest rate of change anticipated for glucose concentration data can be calculated to be between an absolute value of 4 and 5 mg/dL/min [25]. Lowering the lag of the glucose sensor will, in a linear relationship, lower the upper bound on the dynamic measurement error [25]. Since there is a physiological lag between blood glucose levels and ISF glucose levels, it is essential to predict or forecast glucose levels to avoid hyperglycemic incidents caused on by inappropriate insulin administration.

2.2. Existing methods for glucose forecasting

Many approaches which utilize classical machine learning methods such as, such as autoregressive (AR) method, autoregressive exogenous (ARX), auto-regressive moving average (ARMA) [26] and statistical models [27] have been presented in recent years towards glucose forecasting. Furthermore, ML approaches such as random forest (RF) [28], support vector regression (SVR) [29], and grammatical evolution (GE) [30] were extensively used. Shanthi et al. [31] used a traditional statistical approach, an autoregressive integrated moving average (ARIMA) architecture [32], to forecast glucose concentrations taking prediction horizon of 15, 30, 60 minutes. Deep learning approaches have recently been shown to outperform traditional methods. This is owing to the ability of the algorithm to automatically extract key characteristics from training data in order to effectively forecast blood glucose levels. One of the primary problems in developing systems with traditional RNNs is their poor potential to learn long-term dependencies and issues of

exploding or vanishing gradient problems [33]. For blood glucose prediction, an efficient machine learning approaches were presented by Daskalaki et al., [34] who employed an RNN supplied with both glucose and insulin data. Bunesco et al. [35] proposed employing Support Vector Regression model (SVR), taking everyday events such as meal intake and insulin boluses into account. Also, Georga et al. [28] suggested a system for customized glucose prediction that integrated meal information, insulin dosage, exercise model, and the SVR method.

3. Methodology

3.1. System architecture

In a diabetes management system, as shown in Fig. 2, different types of physiological and lifestyle inputs are provided to the glucose forecasting model. The physiological input consists of glucose levels from interstitial fluid, sampled by wearable, continuous glucose monitoring (CGM) sensors. Lifestyle inputs consist of providing the carbohydrate level (CHO). Other lifestyle inputs such as level of physical activity, sleep patterns are extracted from data collected by physical activity wearables. These datapoints are extracted with the help of application programming interfaces (APIs) and the data is stored in the Google Cloud Platform. The AI is trained in the Cloud and deployed in the mobile phone app. In order to validate the proposed deep learning framework, the AI model is validated using clinical datasets.

3.2. Dataset preparation: in silico data

UVA/Padova T1D is used to create the *in silico* dataset used for training and validating the proposed architecture. The simulator provides a stable and well-tested platform for creating simulated scenarios and has been used in previous papers [36,37]. The UVA/Padova T1D, an *in silico* T1D patient simulator, has been approved by the FDA and was used to produce in-silico data. A dataset comprising ten distinct adult T1D patient instances, with 180 days of data, including three daily meals, insulin dosages, and 5-minute interval glucose measurements, totaling 288 daily data points per patient, was generated for personalized cluster analysis.

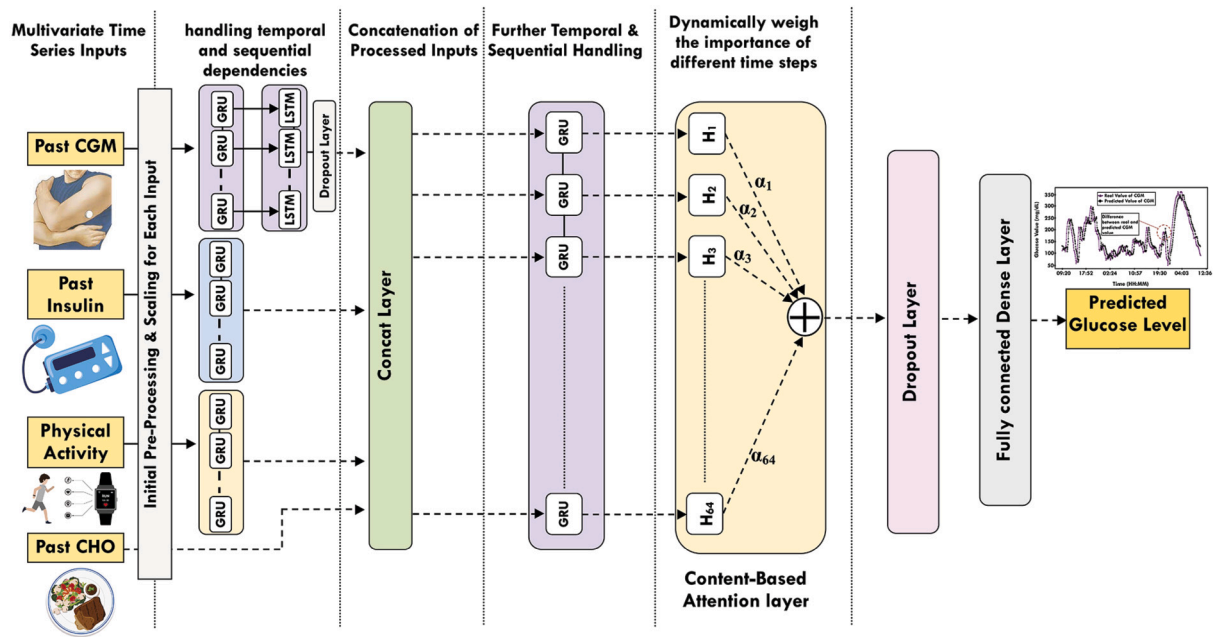


Fig. 3. Fig showed the proposed deep learning model with wide-deep architecture and illustrates the layered design and key components of the model. It is designed for glucose prediction using the input features include PAST CGM, CHO, Insulin and Physical activity.

3.3. Dataset preparation: clinical data

3.3.1. OhioT1DM dataset

The dataset for T1D patients (OhioT1DM) is accessible to the all general public [38]. It comprises data from 12 individuals with Type 1 Diabetes (T1D), collected over an eight-week period. During this time, participants utilized a smartphone app to document their daily activities, coupled with the use of a Medtronic Enlite Continuous Glucose Monitoring (CGM) system and either the company's 530 G or 630 G insulin pumps. Some participants also wore Basis Peak or Empatica Embrace bands to monitor vital signs. The dataset has already been divided into training and testing sets, constituting approximately 80% and 20% of the total samples, respectively. The information for the first six participants in the Blood Glucose Level Prediction (BGLP) Challenge was made available in 2018, and additional data for six participants in the BGLP Challenge 2, incorporating acceleration data, was released in 2020. For our analysis, CGM, CHO, Insulin (include both bolus insulin and basal insulin), Heart Rate, and Steps as input parameters from the 2018 dataset, and acceleration from 2020 dataset was selected.

3.3.2. T2DM dataset

Mobile app and cloud platform development for clinical data collection The architecture consists of Cloud-based data storage for patient data. Fig. 2 illustrates an overall architecture of cloud based patient monitoring system for closed loop diabetes management with incorporation of physical activity. The mobile app collects CGM, meal, insulin and physical activity data through APIs from devices, sending it to the Cloud via our developed APIs. Physical activity data from wearables syncs with a third-party server. Users grant data access through the app after registering on the Data Administration Server. Details of the App are presented in the sections below.

The health and biomarkers of registered patients can be monitored by a Doctor through a Doctor WebApp or mobile app, accessing the server to retrieve patient-specific data for recommendations. The Doctor apps, patient app, and algorithms for glucose forecast and insulin bolus calculation are exclusively developed on Google Cloud. Android apps for patients, displaying CHO, CGM, and physical activity data collected through diverse sensors, store this data in cloud storage. In the cloud, a deep learning model has been deployed for glucose forecast, integrating

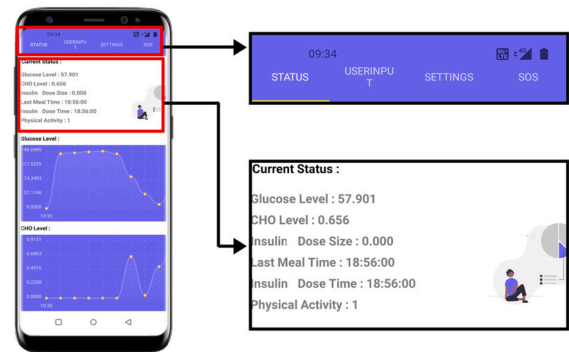


Fig. 4. This figure shows the patient mobile app interface and showing the status window.

previous CGM, CHO, and insulin values with physical activity. Two web interfaces have been specifically developed for Doctors and Patients on the cloud server.

The patient app has been developed using React Native Javascript framework. It consists of a Graphical User Interface (GUI) to display data and functionality for user to send data to the server as shown in Fig. 4. The App communicates with the server using the Representational State Transfer (REST) API.

In this work, the App has been designed to sync to two different Physical Activity monitoring systems, namely GoogleFit and Fitbit. GoogleFit is a physical activity tracking platform designed to collect data from multiple physical activity monitoring devices by different manufacturers and all the connection steps are shown in Fig. 5. Also for CHO calculation we have integrated our app with an image to CHO conversion tab where the participant can enter the food images and from those images we have calculated the carbohydrates quantity with that time stamps as shown in Fig S7 and stored all the datas on our cloud platform including insulin values for prediction of glucose.

Clinical data collection with physical activity: The primary training and validation of the proposed deep learning framework was conducted on OhioT1DM platform. For real-time validation of the framework and platform, one T2D with diabetes subject was monitored and data was

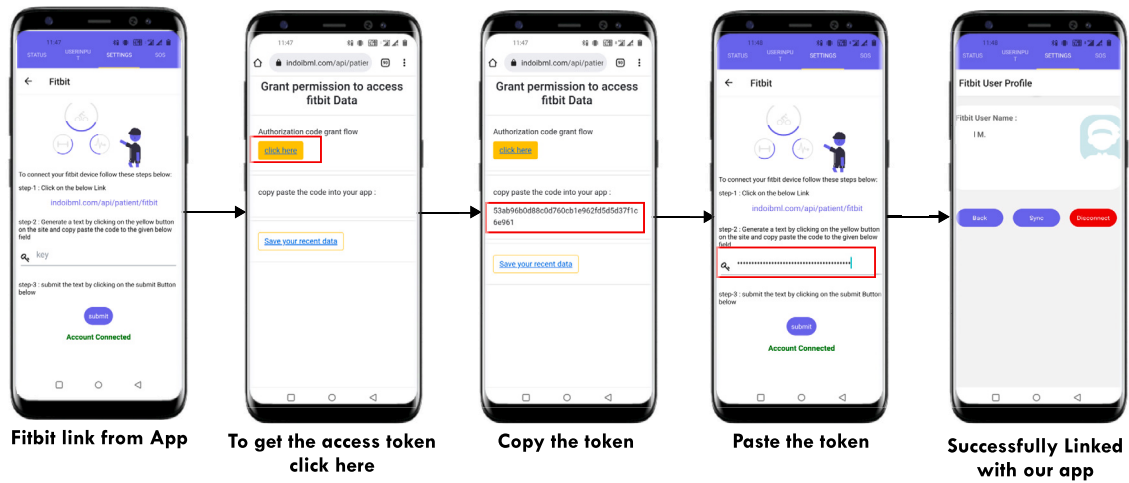


Fig. 5. This figure shows how we integrated our app with Fitbit and retrieved physical activity data from Fitbit.

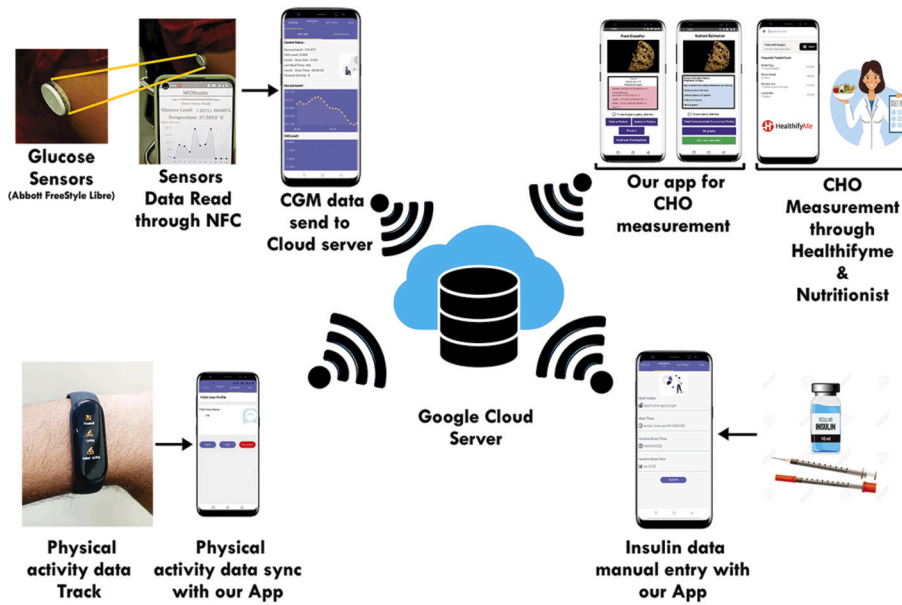


Fig. 6. This figure shows the various sensors and platforms involved for collecting T2D/T1D patient data through the proposed platform.

collected for 14 consecutive days with Freestyle Libre sensor. For this work, ethical clearance taken from Institute Ethics Committee of National Institute of Technology, Rourkela India and SCB Medical College & Hospital, Cuttack, India.

The Abbott Freestyle Libre CGM Sensor was used to build this dataset, and the HealthyMe Android app and our indigenous app was used to record meal information and provide the grams of carbohydrate content with approval of dietitians and nutritionists consultant as shown in the work by Amiri et al. and Sajith et al. [39,40]. Through the fitbit band, we have synced the information on physical activity and physical activity is included in the dataset like physical activity index based on its intensity and heart rate as done in [41]. In our locally collected clinical dataset both of our patients are treated with the Gluconorm G2 tablet daily. This is a combination of Glimepiride and Metformin, increases the insulin secretion in individuals with Type 2 diabetes, for treatment of insulin resistance. The interval between the two samples is 15 minutes according to the freestyle libre sensor. We have generated 96 datapoints in a day, and we stored those information in our cloud server and including insulin values with different time stamps. The Fig. 6 shows how we have collected the T2D/T1D patient data through our developed platform and stored in Google Cloud platform.

3.3.3. Hybrid wide & deep LSTM-GRU model

In order to address the issue of ordinary RNN gradients exploding, Hochreiter and Schmidhuber [42] created deep recurrent neural networks (LSTM NNs). In comparison to an LSTM, GRU uses fewer training parameters, hence it uses less memory during processing and training [43]. GRU layers train more quickly compared to LSTM layers. To take use of the advantages of both types of layers, we have merged GRU and LSTM layers in the suggested network.

In this proposed network architecture, a wide-deep hybrid approach has been implemented, employing separate datapaths—referred to as *wide* and *deep*—to handle distinct input features. The input set consists of Past CGM, CHO, insulin dosage, and physical activity levels from the preceding 8 days, and prior to integration, the data is scaled using Min-Max Scalar. To facilitate efficient dataset creation and enhance modeling of input-output dependencies, a time series generator is utilized.

The network architecture is designed for glucose level forecasting, offering prediction horizons of 15, 30, and 60 minutes—commonly applied in current glucose prediction scenarios. Fig. 3 provides both an overview and detailed representation of the model's structure.

A meticulous breakdown of the architecture reveals a series of processing steps for each input type. Past CGM data undergoes sequential

Table 1

Results of glucose forecasting evaluated on the OhioT1DM subjects (2018 and 2020, clinical dataset) implemented with proposed deep learning framework in terms of MARD, MAE, and RMSE (PH = 15, 30 and 60 Mins).

	Without Physical Activity Inputs: CGM, CHO, Insulin			With Physical Activity Inputs: CGM, CHO, Insulin, Heartrate, Steps		
	RMSE (mg/dL)	MAE (mg/dL)	MARD (%)	RMSE (mg/dL)	MAE (mg/dL)	MARD (%)
Ohio T1D Subjects (Clinical Dataset, 2018)	Prediction Horizon = 15 Min					
	6.18±1.20	4.33±0.66	2.4±0.42	6.01±1.09	4.31±0.60	2.33±0.31
	Prediction Horizon = 30 Min					
	17.26±3.15	12.71±2.23	7.16±1.77	17.19±3.22	12.58±2.34	7.14±1.76
	Prediction Horizon = 60 Min					
	29.63±6.16	22.25±4.83	12.58±3.36	29.41±5.92	21.96±4.67	12.33±3.15
Ohio T1D Subjects (Clinical Dataset, 2020)	Without Physical Activity Inputs: CGM, CHO, Insulin			With Physical Activity Inputs: CGM, CHO, Insulin, Acceleration		
	RMSE (mg/dL)	MAE (mg/dL)	MARD (%)	RMSE (mg/dL)	MAE (mg/dL)	MARD (%)
	Prediction Horizon = 15 Min					
	7.02±1.03	5.28±0.90	3.20±0.46	6.72±0.96	5.05±0.96	3.08±0.44
	Prediction Horizon = 30 Min					
	18.64±2.67	13.46±1.80	7.97±1.35	18.36±2.66	13.14±1.79	7.65±1.31
	Prediction Horizon = 60 Min					
	30.13±3.13	22.20±2.94	13.12±2.39	29.82±3.32	22.02±3.06	12.95±2.40

transformations, passing through a Gated Recurrent Unit (GRU) layer with 128 neurons, followed by a Long Short-Term Memory (LSTM) layer with 128 neurons. Concurrently, CHO input is directly concatenated with the output from the CGM network via a concat layer. The insulin input and physical activity parameter like (heart rate, steps etc.) take a distinct route, traversing another GRU layer with 128 neurons, aimed at handling temporal and sequential dependencies. All the input paths (CGM, CHO, Insulin, physical activity) are subsequently merged and collectively fed into a GRU layer with 64 neurons. The predicted CGM value emerges as the output from the penultimate GRU layer.

An innovative addition to this architecture is the incorporation of a Content-Based Attention Mechanism within the GRU layer. This dynamic mechanism assigns varying weights to different time steps based on the content of the input sequence, augmenting the model's ability to focus on pertinent information during predictions.

The strategic choice to combine Gated Recurrent Units (GRU) and Long Short-Term Memory (LSTM) layers capitalizes on the strengths of each. The simplicity and faster training of GRU complement the memory management capabilities of LSTM, resulting in a more effective and efficient neural network for glucose forecasting, particularly when dealing with a reduced amount of input data.

The adaptive nature of the architecture is highlighted in its response to varying data availability for different inputs. Limited CHO data, with only 3 points daily, is directly concatenated with the past CGM input. Conversely, for data-rich inputs like physical activity or insulin, a dedicated GRU layer is employed before merging with CGM, accommodating the increased number of data points. This proposed model seamlessly integrates a wide-deep hybrid structure, attention mechanisms, and a judicious combination of GRU and LSTM layers. The accompanying figure visually encapsulates these components, underscoring the model's proficiency in capturing dependencies, processing sequential features, and ultimately enhancing predictive accuracy for personalized glucose forecasting.

4. Results

The presented novel neural network framework's performance is analyzed and compared to the current state of the art. The performance of the algorithms is estimated from values of mean absolute error (MAE),

more importantly mean absolute relative difference (MARD) and finally root mean square error (RMSE) across forecast window of 15, 30, and 60-minute. MARD is a performance metric widely used for this application [25]. In addition to the model performance, the quantity of training data needed for various standard model architectures and architectures proposed in this paper, was evaluated.

4.1. Performance metrics

MARD is measured as a %, RMSE is measured in mg/dL of glucose concentration, and the mean absolute error (MAE) in mg/dL of glucose concentration. They were utilized to compute the effectiveness and precision of the suggested method. The detailed mathematical formulae for the different performance metrics are as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum_{k=1}^N (y_k - y'_k)^2} \quad (6)$$

$$MARD = \frac{1}{N} \sum_{k=1}^N \frac{|y_k - y'_k|}{y_k} \quad (7)$$

$$MAE = \frac{1}{N} \sum_{k=1}^N |y_k - y'_k| \quad (8)$$

where y_k is the actual or original glucose reading for k_{th} instance, y'_k denotes the forecasted value for k_{th} instance, and the total number of data points is denoted by N. The MARD, RMSE, and MAE give us a sense of how well we can forecast the results. One of the benefits of glucose forecasting, as previously indicated, is the avoidance of unfavourable glycaemic events. The above three criteria were chosen because they are often used to evaluate glucose predictions in the literature. As explained in the study L. Heinemann et al. [44], The common factor in glucose prediction is MARD.

4.2. MARD, RMSE & MAE from Ohio T1D patients and T2D clinical dataset

The results of MARD, RMSE and MAE for the prediction of glucose levels are summarised in Table 1 for OhioT1D dataset and detailed

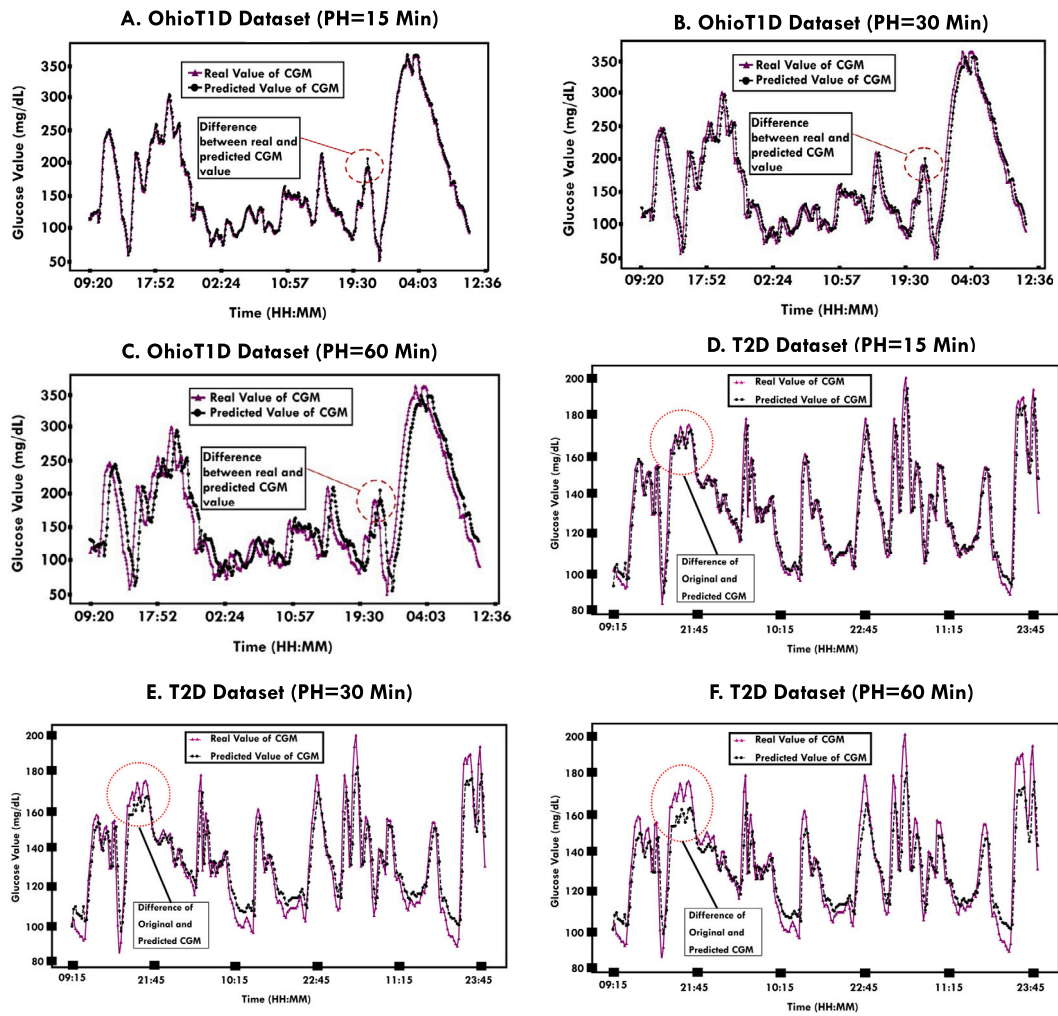


Fig. 7. This figure shows the predicted and actual CGM values for comparison, using the proposed network for 15, 30 and 60 min PH on clinical Dataset OhioT1DM subject (Patient ID 559) and for T2D clinical dataset subject (Patient ID P1810M2701).

results are showed in Table S4-S7 in supplementary material. The algorithms' efficacy has been mentioned in the above tables for different PH with different input parameters. In the Fig. 7, the difference between actual and predicted glucose values for the proposed architecture is shown, for a single OhioT1DM subject (Patient ID 559). Here, the difference between predicted and actual value of glucose and time lag is very minimal for proposed architecture. In addition, the proposed model provides relatively improved forecasting accuracy during both hyperglycaemic and hypoglycaemic phases.

Similarly Table 2 shows difference between predicted and actual value of glucose for T2D clinical dataset. The results of MARD, RMSE and MAE for the prediction of glucose levels are summarised in Table 2 for different PH with different input parameters for all T2D subjects. Similarly Fig. 7 (D, E and F) shows difference between predicted and actual value of glucose for T2D clinical dataset (Patient ID P1810M2701).

4.3. MARD, RMSE & MAE from *in silico* patients

The above architecture that was mentioned WNet was further verified using an *in silico* T1D dataset for 10 adult patients, and all detailed results are shown in Fig S1, S2, and S3 in the supplementary material. Additionally, detailed findings for glucose forecasting are provided in supplementary tables S1, S2, and S3 for multiple deep learning models for *in silico* T1D dataset. Existing work comparison table for *in silico*

dataset involving glucose prediction has demonstrated the use of ANN, ARIMA, LSTM, CRNN and BiLSTM networks as shown in Table S10 in supplementary material.

4.4. Clarke Error Grid Analysis on Ohio T1D dataset

Clarke Error Grid analysis is significant because it evaluates not just how accurate a glucose monitoring system is in a statistical sense, but how those accuracy levels translate into real-world clinical decisions. By categorizing potential errors according to their impact on patient care, the Clarke Error Grid analysis provides a comprehensive assessment of a device's suitability for use in managing diabetes [45]. The findings of this examination for three different time intervals, namely 15 minutes, 30 minutes, and 60 minutes, are shown in Fig. 8 and Table 3. The study reveals that, regardless of the pH values, the majority of the predictions for all prediction horizons, as depicted in Table II, fall within region A. This region signifies that the predictions are within a 20% range of the actual value. Only a small number of points fall into region B, representing forecasts that exceed 20% but do not result in any negative treatment effects. Only a very small number of predictions, namely 1-3, fall into area C or other regions. These predictions suggest that these points may result in unnecessary treatment, but they are not likely to be potentially dangerous.

Table 2
Results of glucose forecasting for T2D clinical dataset with physical activity and without physical activity implemented with the proposed deep learning framework in terms of MARD, MAE, and RMSE (PH = 15, 30 and 60 Mins).

Patient ID	RMSE (mg/dL)	MAE (mg/dL)	MARD (%)	RMSE (mg/dL)	MAE (mg/dL)	MARD (%)
Prediction Horizon = 15 Min						
Without Physical Activity (Inputs: CGM, CHO, Insulin)			With Physical Activity (Inputs: CGM, CHO, Insulin, Heart rate, Steps)			
Adult 1 (P1810M2701)	8.91	6.75	4.03	8.31	6.56	3.84
Adult 2 (P1905F5501)	7.15	5.73	3.43	6.81	5.53	2.91
Avg±SD	8.03±0.88	6.24±0.51	3.73±0.30	7.56±0.75	6.04±0.51	3.37±0.46
Prediction Horizon = 30 Min						
Adult 1 (P1810M2701)	20.61	17.50	9.65	20.44	17.29	9.46
Adult 2 (P1905F5501)	19.23	14.37	8.23	19.14	14.09	8.17
Avg±SD	19.92±0.69	15.93±1.56	8.94±0.71	19.79±0.65	15.69±1.60	8.81±0.64
Prediction Horizon = 60 Min						
Adult 1 (P1810M2701)	32.71	23.13	19.95	32.28	24.70	18.27
Adult 2 (P1905F5501)	31.49	22.45	14.47	30.63	21.37	13.51
Avg±SD	32.10±0.61	22.79±0.34	17.21±2.74	31.45±0.82	23.03±1.66	15.89±2.38

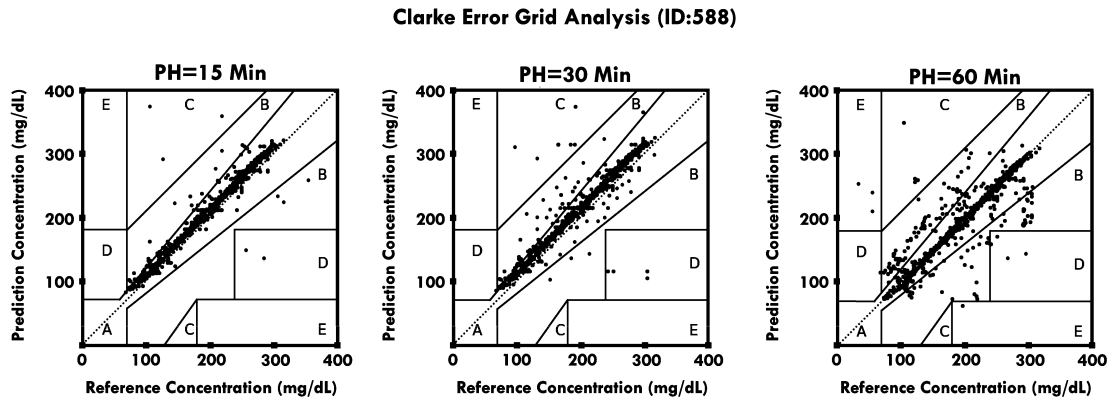


Fig. 8. Visual representation of the predicted CGM data and reference CGM data Clarke Error Grid Analysis was performed on the OhioT1DM Dataset (ID:588) in order to make predictions on a 15, 30, and 60 PH.

Table 3
Analysis of the Clarke Error Grid for different prediction horizons.

Patient ID	Ohio Dataset, 2018 With PA (CGM, Insulin, CHO, Heart Rate, Steps)														
	PH = 15 Min					PH = 30 Min					PH = 60 Min				
	Zone A (%)	Zone B (%)	Zone C (%)	Zone D (%)	Zone E (%)	Zone A (%)	Zone B (%)	Zone C (%)	Zone D (%)	Zone E (%)	Zone A (%)	Zone B (%)	Zone C (%)	Zone D (%)	Zone E (%)
559	95.60	3.80	0.60	0.00	0.00	91.35	7.98	0.51	0.16	0.00	85.63	13.72	0.15	0.50	0.00
563	96.74	2.89	0.37	0.00	0.00	90.89	8.55	0.44	0.12	0.00	84.75	14.47	0.53	0.19	0.06
570	98.09	1.75	0.16	0.00	0.00	93.15	6.33	0.51	0.01	0.00	88.66	11.03	0.23	0.08	0.00
575	94.85	4.73	0.37	0.05	0.00	89.73	9.59	0.45	0.12	0.11	77.89	20.12	1.47	0.29	0.23
588	96.56	2.78	0.58	0.08	0.00	92.26	7.08	0.43	0.23	0.00	83.73	15.47	0.61	0.12	0.07
591	93.88	4.96	0.89	0.27	0.00	89.19	9.99	0.59	0.23	0.00	81.87	16.51	0.89	0.67	0.06
Avg.	95.95	3.48	0.49	0.06	0.00	91.09	8.25	0.48	0.14	0.01	83.75	15.22	0.64	0.30	0.07
	Ohio Dataset, 2020 With PA (CGM, Insulin, CHO, Acceleration)														
	PH = 15 Min					PH = 30 Min					PH = 60 Min				
	Zone A (%)	Zone B (%)	Zone C (%)	Zone D (%)	Zone E (%)	Zone A (%)	Zone B (%)	Zone C (%)	Zone D (%)	Zone E (%)	Zone A (%)	Zone B (%)	Zone C (%)	Zone D (%)	Zone E (%)
540	96.74	3.05	0.17	0.04	0.00	90.92	7.98	0.67	0.43	0.00	85.32	13.95	0.35	0.29	0.09
544	95.27	4.19	0.51	0.03	0.00	92.01	7.85	0.12	0.02	0.00	86.32	12.44	0.85	0.31	0.08
552	94.37	5.02	0.58	0.03	0.00	88.76	10.87	0.33	0.03	0.01	83.78	14.67	0.98	0.45	0.12
567	95.37	4.13	0.37	0.13	0.00	89.01	10.29	0.41	0.23	0.06	84.93	14.39	0.65	0.03	0.00
584	92.95	6.05	0.68	0.29	0.03	89.61	9.78	0.47	0.12	0.02	81.78	16.83	0.99	0.31	0.09
596	94.89	4.02	0.83	0.19	0.07	89.97	9.83	0.15	0.03	0.02	82.79	15.98	0.89	0.23	0.11
Avg.	94.93	4.41	0.52	0.11	0.01	90.04	9.43	0.35	0.14	0.01	84.15	14.71	0.78	0.27	0.08

Table 4Comparisons of forecast performance with different existing framework for PH = 15 min, 30 min, 60 min on *Clinical* dataset.

SL No.	Related Existing Work	Publication Year	Architecture Utilized	RMSE (in mg/dL)	MAE (in mg/dL)	MARD (in %)	Time Lag (in Min)
Prediction Time Horizon (PH = 15 min)							
1	Zecchin et al. [46]	2016	NN	14.1	-	-	-
2	A Muhammad et al. [47]	2021	Nonlinear ANN	7.81	-	-	-
3	Proposed Work	2024	This paper	6.01±1.09	4.31±0.60	2.33±0.31	5.0
Prediction Time Horizon (PH = 30 min)							
4	Li et al. [48]	2019	CRNN	21.07±2.35	-	11.61±2.18	19.3±3.1
5	Zhu et al. [15]	2020	DRNN	18.9±2.6	-	8.7±2.2	8.41±4.1
6	Li et al. [49]	2020	GluNET	19.28±2.76	-	8.73±1.78	8.03±4.07
7	Dudukcu et al. [50]	2021	LSTM+WaveNet+GRU	21.90	15.87	-	-
9	Daniels et al. [51]	2022	CRNN-MTL	18.8±2.3	13.2±1.6	-	13.8±5.1
10	Daniels et al. [51]	2022	CRNN-MTL-GV	18.8±2.8	13.2±1.5	-	13.3±5.9
8	Zhu et al. [52]	2023	FCN and Meta	18.64±2.60	13.25±1.67	-	6.75±4.30
12	G Yang et al. [53]	2023	PBGTAM	20.57±2.16	-	8.68±1.50	-
11	Proposed Work	2024	This paper	17.19±3.22	12.58±2.34	7.14±1.76	20.0
Prediction Time Horizon (PH = 60 min)							
12	Li et al. [48]	2019	CRNN	33.27±4.79	-	19.01±4.46	29.3±9.4
13	Li et al. [49]	2020	GluNET	31.83±3.49	-	16.11±4.49	17.78±8.00
7	Dudukcu et al. [50]	2021	LSTM+WaveNet+GRU	35.10	26.41	-	-
15	Daniels et al. [51]	2022	CRNN-MTL	31.8±3.9	23.4±3.0	-	20.4±8.3
16	Daniels et al. [51]	2022	CRNN-MTL-GV	32.3±3.9	23.6±3	-	17.5±8.8
14	Zhu et al. [52]	2023	FCN and Meta	31.07±3.62	22.86±2.89	-	14.58±9.91
19	G Yang et al. [53]	2023	PBGTAM	35.09±3.00	-	15.27±2.69	-
17	Proposed Work	2024	This paper	29.41±5.92	21.96±4.67	12.33±3.15	40.0

4.5. Training time

Loss curves provide an overview of the training process as well as the network's learning trajectory. The training time can be extracted from the loss curves. Fig. S1, S2, S3, S4, S5 and S6 in supplementary material show training and validation loss curves for different models in case of *in silico* dataset and *T1D Ohio T1DM* and *T2D clinical* dataset. For the proposed architecture, a single epoch takes only 1 sec and the training completes within 10 seconds maximum.

4.6. Personalisation of the deep learning framework

The performance of the deep learning framework was tested for personalisation by training and testing in two different ways. Firstly, the framework was trained on 8 days of clinical data followed by testing on different test sets 1-5. Each test set consist 3 days of data with each set being temporally distant from the training set. For *personalisation*, the same framework was trained, however, the training data was selected on days just prior to those included in the test sets. The results are shown in Fig. 9.

5. Discussion

In this paper, glucose prediction was considered as a time series prediction task. RNNs, LSTMs, and GRUs have been generally used for time series prediction. However no hybrid architectures where both LSTM and GRU, with a wide-deep approach, have been explored and studied earlier. The performance metrics involve RMSE, MAE, MARD, and time lag. The proposed platform presented in the paper was evaluated for feasibility and applicability towards a real-life scenario. Extensive statistical analysis could not be performed as some previous works have not reported standard deviation of the performance metrics. Detailed statistical analysis is presented in Supplementary S11 and S12.

In this paper, an effort was made to combine the benefits of both LSTM and GRU layers by implementing a hybrid network in wide-deep manner. Simultaneously, attempt was also made to reduce the quantity

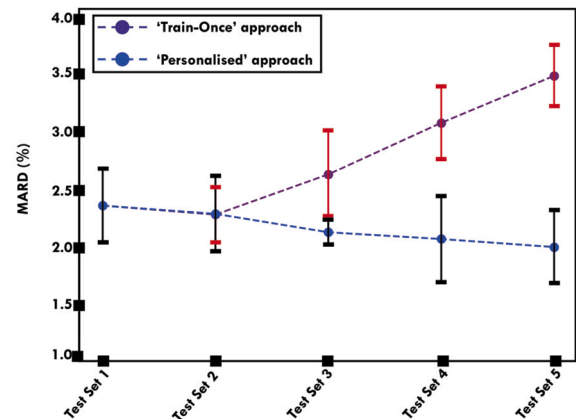


Fig. 9. This figure shows training for personalised (blue line) approach and training once approach (purple line). In personalised approach test set and train set are shifted and trained once approach training is done only for first 8 days and test dataset is shifted.

of training data required, hence improving *personalisation* i.e. the model is trained on more recent data and represents any recent physiological or pathological changes better. This also reduces the training time of the model.

In the proposed model, the input feature *CGM* is initially processed through a set of GRU layers followed by LSTM layers. The GRU layers were initially added for simplicity compared to LSTM layers. However, it was observed that after adding several GRU layers in the beginning, if LSTM layers are added after that, then the performance metrics such as MARD and RMSE improve. This indicates that for input feature *CGM*, at later stages it is crucial to preserve the *long-term* state of the *short-term* states of the inputs. This is possible using the LSTM layers. In the dataset, the number of *CGM* datapoints also greatly outnumber those of *CHO* and *Insulin* datapoints. Hence, they require a more complex pathway compared to other input features. *CHO* and *Insulin* inputs require only GRU layers with *insulin* requiring more GRU layers than *CHO* input. We

observer that *Insulin* and *Physical Activity (PA)* inputs require similar layers for optimal performance.

In clinical validation on the Ohio T1DM Dataset, the RMSE for 15-minute prediction horizon is 6.01 ± 1.09 mg/dL, which is the best among existing approaches. In addition, as compared to prior research, we achieved higher MAE (4.31 ± 0.60 mg/dL) and MARD ($2.33 \pm 0.31\%$) for the 15-minute forecast horizon.

Similarly, this proposed design is provided RMSE of 17.19 ± 3.22 mg/dL, MAE of 12.58 ± 2.34 mg/dL, and MARD of $7.14 \pm 1.76\%$ across PH = 30. For PH = 60-minute, the RMSE was determined to be 29.41 ± 5.92 mg/dL, MAE is 21.96 ± 4.67 mg/dL, and MARD is $12.33 \pm 3.15\%$ for the proposed framework.

The RMSE for the 15 min prediction horizon for the WNet network utilizing the T2D clinical dataset with physical activity is 7.56 ± 0.75 mg/dL. In addition, we obtained superior MAE 6.04 ± 0.51 mg/dL and MARD $3.37 \pm 0.46\%$ for the 15-minute forecast horizon when compared to previous studies. Similarly, over a 30 minute prediction horizon, this design produced RMSE as 19.79 ± 0.65 mg/dL, MAE as 15.69 ± 1.60 mg/dL, and MARD as $8.81 \pm 0.64\%$. We discovered that the RMSE is 31.45 ± 0.82 mg/dL, MAE is 23.03 ± 1.66 mg/dL, and MARD is $15.89 \pm 2.38\%$ for a 60-minute forecast horizon for the proposed WNet framework. Similarly, this proposed design is provided when physical activity is not considered RMSE of 8.03 ± 0.88 mg/dL, MAE of 6.24 ± 0.51 mg/dL, and MARD of $3.73 \pm 0.30\%$ across PH = 15. For PH = 30-minute, the RMSE was determined to be 19.92 ± 0.69 mg/dL, MAE is 15.93 ± 1.56 mg/dL, and MARD is $8.94 \pm 0.71\%$. Similarly, for PH = 60 min RMSE is 32.10 ± 0.61 mg/dL, and MAE is 22.79 ± 0.34 mg/dL and MARD is $17.21 \pm 2.74\%$. Table 4 shows comparisons of forecast performance with different existing frameworks for PH = 15 min, 30 min, and 60 min on the Clinical dataset.

We also observe in clinical datasets, *personalisation* of the framework by training on data which are less temporally distant data leads to better performance of the model as evidenced through lower MARD compared to training only once temporally distant datasets. This also indicates that it is required to update the AI models frequently for better performance. This is observed when the MARD reduces with data which is trained progressively.

Additionally, we also present a cloud-based platform which can be used to record patient data both from CGM sensors and physical activity monitoring devices seamlessly to record data from both sensors. This data is then used by the model deployed in the mobile phone app to perform glucose predictions. The complete platform was validated using clinical data ranging from data collection, data integration and glucose forecasting in individual with Type 2 diabetes. The Google Cloud based platform was also tested for handling user load.

To summarise, as observed from the results, the deep learning framework offers several benefits. The training dataset utilizes only 8 days of data compared to other existing work which is trained on almost 6 months of previous data. Similarly for T1D and T2D clinical datasets, we are training our framework with 8 days of data and get good accuracy. This also helps in training the algorithm to better reflect the changing physiological and pathological condition of the patient. The performance of proposed deep learning framework can be attributed to two factors i.e. it combines the benefits of LSTM and GRU layers. In addition, it better models the inter-relationships between inputs and outputs by applying different layers and pathways to different inputs.

The paper is primarily focuses on the novel aspects of the platform i.e. model architecture and validation. Regarding the Cloud and App aspects of the platform, we have provided the overall system architecture. The model's initial results are encouraging, but more research is needed to improve its efficacy and applicability. This validation is confined to a small clinical dataset, hence more clinical trials are needed to generalise the model to a larger patient population. The model has been deployed on a Google cloud server, but edge devices, essential for real-time applications, are still being improved. Data collecting issues

such sensor malfunctions and transmission errors limit the model's robustness, which needs further investigation.

6. Conclusion

In this work, the use of hybrid network such as the proposed one for glucose prediction results in improved performance compared to existing DNN architectures used for glucose prediction. The model also demonstrates the benefits of using a wide-deep approach for different features. The advantages of using a wide-deep approach are not diminished even if the amount of training data is reduced when compared to previous works. Furthermore, a personalised approach relying on smaller, more recent data leads to better forecasting accuracy. Future work may consist of implementing the model on edge hardware.

Ethical approval

Ethical clearance for this work was obtained from the Institute Ethics Committee of National Institute of Technology, Rourkela, India (Certificate Number: NITR/IEC/2022/M/09, Issued Date: 9/06/2022), and SCB Medical College & Hospital, Cuttack, India (IEC Application Number: 989, Date: 4/02/2022).

Funding

The authors also acknowledge support from Indian Council of Medical Research under grant no. IIRP/23-24/7414 and Institute Fellowship (PhD), NIT Rourkela.

Declaration of competing interest

None.

Acknowledgement

The authors would also like to thank Subhrajyoti Sankar Sau, former undergraduate student at NIT Rourkela, for his assistance in developing the App and Cloud Platform as part of his studies at NIT Rourkela.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.medengphy.2024.104241>.

References

- [1] Facchinetti A, Favero S, Sparacino G, Cobelli C. An online failure detection method of the glucose sensor-insulin pump system: improved overnight safety of type-1 diabetic subjects. *IEEE Trans Biomed Eng* 2012;60:406–16.
- [2] Zavitsanou S, Mantalaris A, Georgiadis MC, Pistikopoulos EN. In silico closed-loop control validation studies for optimal insulin delivery in type 1 diabetes. *IEEE Trans Biomed Eng* 2015;62:2369–78.
- [3] Vettoretti M, Facchinetti A, Sparacino G, Cobelli C. Type-1 diabetes patient decision simulator for in silico testing safety and effectiveness of insulin treatments. *IEEE Trans Biomed Eng* 2017;65:1281–90.
- [4] Kiani S, Rezaei P, Fakhr M. Dual-frequency microwave resonant sensor to detect noninvasive glucose-level changes through the fingertip. *IEEE Trans Instrum Meas* 2021;70:1–8.
- [5] Omer AE, Hojjati-Firoozabadi A, Gigoyan S, Safavi-Naeini S, Shaker G. Non-reciprocal whispering-gallery-mode resonator for sensitive blood glucose monitoring. *IEEE Trans Instrum Meas* 2022;71:1–12.
- [6] Facchinetti A. Continuous glucose monitoring sensors: past, present and future algorithmic challenges. *Sensors* 2016;16:2093.
- [7] Syafie S, AlHarbi F, Alshehri AA, Hasanain B. Pid and lqg controllers for diabetes system with internal delay: a comparison study. *Biomed Phys Eng Express* 2023;9:035031.
- [8] Oviedo S, Vehí J, Calm R, Armengol J. A review of personalized blood glucose prediction strategies for t1dm patients. *Int J Numer Methods Biomed Eng* 2017;33:e2833.
- [9] Pesl P, Herrero P, Reddy M, Xenou M, Oliver N, Johnston D, et al. An advanced bolus calculator for type 1 diabetes: system architecture and usability results. *IEEE J Biomed Health Inform* 2015;20:11–7.

- [10] Grant P. A new approach to diabetic control: fuzzy logic and insulin pump technology. *Med Eng Phys* 2007;29:824–7.
- [11] Zhu T, Li K, Herrero P, Chen J, Georgiou P. A deep learning algorithm for personalized blood glucose prediction. In: *KHD@ IJCAI*; 2018. p. 64–78.
- [12] Sun Q, Jankovic MV, Bally L, Mougiakakou SG. Predicting blood glucose with an lstm and bi-lstm based deep neural network. In: *2018 14th symposium on neural networks and applications (NEUREL)*. IEEE; 2018. p. 1–5.
- [13] Mohebbi A, Johansen AR, Hansen N, Christensen PE, Tarp JM, Jensen ML, et al. Short term blood glucose prediction based on continuous glucose monitoring data. In: *2020 42nd annual international conference of the IEEE engineering in medicine & biology society (EMBC)*. IEEE; 2020. p. 5140–5.
- [14] Ali JB, Hamdi T, Fnaiech N, Di Costanzo V, Fnaiech F, Ginoux J-M. Continuous blood glucose level prediction of type 1 diabetes based on artificial neural network. *Biocybern Biomed Eng* 2018;38:828–40.
- [15] Freiburghaus J, Rizzotti-Kaddouri A, Albertetti F. A deep learning approach for blood glucose prediction of type 1 diabetes. In: *CEUR workshop proceedings*, vol. 2675. 2020. p. 131–5.
- [16] Zhu T, Kuang L, Li K, Zeng J, Herrero P, Georgiou P. Blood glucose prediction in type 1 diabetes using deep learning on the edge. In: *2021 IEEE international symposium on circuits and systems (ISCAS)*. IEEE; 2021. p. 1–5.
- [17] Maweu BM, Shamsuddin R, Dakshit S, Prabhakaran B. Generating healthcare time series data for improving diagnostic accuracy of deep neural networks. *IEEE Trans Instrum Meas* 2021;70:1–15.
- [18] Mu G, Chen J. Developing a conditional variational autoencoder to guide spectral data augmentation for calibration modeling. *IEEE Trans Instrum Meas* 2022;71:1–8.
- [19] Wang C-C, Yang C-H, Wang C-S, Xu D, Huang B-S. Artificial neural networks in the selection of shoe lasts for people with mild diabetes. *Med Eng Phys* 2019;64:37–45.
- [20] Pai PP, De A, Banerjee S. Accuracy enhancement for noninvasive glucose estimation using dual-wavelength photoacoustic measurements and kernel-based calibration. *IEEE Trans Instrum Meas* 2017;67:126–36.
- [21] Kalita D, Sharma H, Mirza KB. Continuous glucose, insulin and lifestyle data augmentation in artificial pancreas using adaptive generative and discriminative models. *IEEE J Biomed Health Inform* 2024.
- [22] Zarkogianni K, Mitsis K, Litsa E, Arredondo M-T, Fico G, Fioravanti A, et al. Comparative assessment of glucose prediction models for patients with type 1 diabetes mellitus using sensors for glucose and physical activity monitoring. *Med Biol Eng Comput* 2015;53:1333–43.
- [23] Mhaskar HN, Pereverzyev SV, Van der Walt MD. A deep learning approach to diabetic blood glucose prediction. *Front Appl Math Stat* 2017;3:14.
- [24] Kalita D, Mirza KB. Glucose prediction using wide-deep lstm network for accurate insulin dosing in artificial pancreas. In: *2022 44th annual international conference of the IEEE engineering in medicine and biology society*. IEEE; 2022. p. 4426–9.
- [25] Huyett LM, Dassau E, Zisser HC, Doyle FJ. Glucose sensor dynamics and the artificial pancreas: the impact of lag on sensor measurement and controller performance. *IEEE Control Syst Mag* 2018;38:30–46.
- [26] Krizhevsky A, Sutskever I, Hinton GE. Imagenet classification with deep convolutional neural networks. *Adv Neural Inf Process Syst* 2012;25.
- [27] Zecchin C, Facchinetti A, Sparacino G, De Nicolao G, Cobelli C. Neural network incorporating meal information improves accuracy of short-time prediction of glucose concentration. *IEEE Trans Biomed Eng* 2012;59:1550–60.
- [28] Georga EI, Protopappas VC, Ardigo D, Marina M, Zavaroni I, Polyzos D, et al. Multivariate prediction of subcutaneous glucose concentration in type 1 diabetes patients based on support vector regression. *IEEE J Biomed Health Inform* 2012;17:71–81.
- [29] Zhao C, Dassau E, Jovanović L, Zisser HC, Doyle III FJ, Seborg DE. Predicting subcutaneous glucose concentration using a latent-variable-based statistical method for type 1 diabetes mellitus. *J Diabetes Sci Technol* 2012;6:617–33.
- [30] Li J, Fernando C. Smartphone-based personalized blood glucose prediction. *ICT Express* 2016;2:150–4.
- [31] Contreras I, Oviedo S, Vettoretti M, Visentin R, Vehí J. Personalized blood glucose prediction: a hybrid approach using grammatical evolution and physiological models. *PLoS ONE* 2017;12:e0187754.
- [32] Shanthy S. Advanced engineering informatics a novel approach for the prediction of glucose concentration in type 1 diabetes ahead in time through arima and differential evolution 2011;38:4182–6.
- [33] Georga EI, Protopappas VC, Fotiadis DI. Glucose prediction in type 1 and type 2 diabetic patients using data driven techniques. In: *Knowledge-oriented applications in data mining*; 2011. p. 277–96.
- [34] Yang J, Li L, Shi Y, Xie X. An arima model with adaptive orders for predicting blood glucose concentrations and hypoglycemia. *IEEE J Biomed Health Inform* 2018;23:1251–60.
- [35] Daskalaki E, Prountzou A, Diem P, Mougiakakou SG. Real-time adaptive models for the personalized prediction of glycemic profile in type 1 diabetes patients. *Diabetes Technol Ther* 2012;14:168–74.
- [36] Man CD, Micheletto F, Lv D, Breton M, Kovatchev B, Cobelli C. The uva/padova type 1 diabetes simulator: new features. *J Diabetes Sci Technol* 2014;8:26–34.
- [37] Herrero P, Pesl P, Bondia J, Reddy M, Oliver N, Georgiou P, et al. Method for automatic adjustment of an insulin bolus calculator: in silico robustness evaluation under intra-day variability. *Comput Methods Programs Biomed* 2015;119:1–8.
- [38] Marling C, Bunesco R. The ohio1dm dataset for blood glucose level prediction. In: *Update 2020*. CEUR workshop proceedings, vol. 2675. NIH Public Access; 2020. p. 71.
- [39] Amiri M, Li J, Hasan W. Personalized flexible meal planning for individuals with diet-related health concerns: system design and feasibility validation study. *JMIR Form Res* 2023;7:e46434.
- [40] Sajith R, Khatua C, Kalita D, Mirza KB. Nutrient estimation from images of food for diet management in diabetic patients. In: *2023 world conference on communication & computing (WCONF)*. IEEE; 2023. p. 1–6.
- [41] Rashid M, Samadi S, Sevil M, Hajizadeh I, Kolodziej P, Hobbs N, et al. Simulation software for assessment of nonlinear and adaptive multivariable control algorithms: glucose-insulin dynamics in type 1 diabetes. *Comput Chem Eng* 2019;130:106565.
- [42] Hochreiter S, Schmidhuber J. Long short-term memory. *Neural Comput* 1997;9:1735–80.
- [43] Almalki AJ, Wocjan P. Forecasting method based upon gru-based deep learning model. In: *2020 international conference on computational science and computational intelligence (CSCI)*. IEEE; 2020. p. 534–8.
- [44] Heinemann L, Schoemaker M, Schmelzeisen-Redecker G, Hinzmann R, Kassab A, Freckmann G, et al. Benefits and limitations of mard as a performance parameter for continuous glucose monitoring in the interstitial space. *J Diabetes Sci Technol* 2020;14:135–50.
- [45] Clarke WL. The original Clarke error grid analysis (ega). *Diabetes Technol Ther* 2005;7:776–9.
- [46] Zecchin C, Facchinetti A, Sparacino G, Cobelli C. How much is short-term glucose prediction in type 1 diabetes improved by adding insulin delivery and meal content information to cgm data? A proof-of-concept study. *J Diabetes Sci Technol* 2016;10:1149–60.
- [47] Asad M, Qamar U, Abbas M. Blood glucose level prediction of diabetic type 1 patients using nonlinear autoregressive neural networks. *J Healthc Eng* 2021;2021:1–7.
- [48] Li K, Daniels J, Liu C, Herrero P, Georgiou P. Convolutional recurrent neural networks for glucose prediction. *IEEE J Biomed Health Inform* 2019;24:603–13.
- [49] Li K, Liu C, Zhu T, Herrero P, Georgiou P. Plunet: a deep learning framework for accurate glucose forecasting. *IEEE J Biomed Health Inform* 2020;24:414–23.
- [50] Dudukcu HV, Taskiran M, Yildirim T. Blood glucose prediction with deep neural networks using weighted decision level fusion. *Biocybern Biomed Eng* 2021;41:1208–23.
- [51] Daniels J, Herrero P, Georgiou P. A multitask learning approach to personalized blood glucose prediction. *IEEE J Biomed Health Inform* 2022;26:436–45.
- [52] Zhu T, Li K, Herrero P, Georgiou P. Personalized blood glucose prediction for type 1 diabetes using evidential deep learning and meta-learning. *IEEE Trans Biomed Eng* 2023;70:193–204.
- [53] Yang G, Liu S, Li Y, He L. Short-term prediction method of blood glucose based on temporal multi-head attention mechanism for diabetic patients. *Biomed Signal Process Control* 2023;82:104552.